

Who holds government debt?

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Abstract

This paper develops a model in which the composition of public debt across domestic banks, domestic non-banks, and foreign investors is determined by endogenous borrowing limits. The government faces default constraints on non-bank creditors and convex costs of financial repression on bank-held debt. These constraints generate threshold effects: when total debt exceeds a critical level, the government exhausts non-bank borrowing capacity and allocates marginal debt to banks.

We structurally estimate the model using maximum likelihood on panel data of debt composition. The results identify distinct borrowing limits across creditor types and provide a tractable framework to interpret cross-country and time variation in who holds government debt.

Keywords Sovereign Debt; Debt Composition; Financial Repression

1 Introduction

Who holds government debt is central to macroeconomic stability. The distribution of debt across domestic banks, domestic non-banks, and foreign investors shapes default incentives, financial repression, and the transmission of fiscal shocks. Yet most models of sovereign debt focus on the total level of debt and treat its composition as exogenous.

This paper develops a simple framework in which the composition of government debt is endogenously determined by borrowing constraints and repression costs. The government faces default limits on domestic and foreign non-bank creditors, while accumulating debt in the banking sector generates convex financial repression costs. These constraints define a no-default region in debt space and imply threshold effects: beyond a critical level of total debt, the government exhausts non-bank borrowing capacity and reallocates marginal debt to banks.

We structurally estimate the model using panel data on debt composition across countries and over time. The estimation identifies distinct borrowing limits across creditor types and provides a tractable framework to interpret variation in who holds government debt.

2 Related literature and organization

2.1 Related literature

Our paper builds on, and brings together, four strands of the sovereign-debt literature: work on the composition of debt holders and default risk; the financial-repression literature on domestic debt bias; the sovereign-bank “doom loop”; and the classical literature on the optimal level of government debt. Existing work addresses pieces of the picture: the default literature studies the choice between domestic and foreign creditors; the financial-repression literature explains why governments push debt onto banks; and the home-bias literature documents preferences for domestic sovereign debt. Our contribution is to shift the focus from the *level* of debt to the *allocation* across holder types and to provide a structural framework that links observable financial-sector heterogeneity to the cross-country pattern of debt composition.

2.1.1 Government debt composition and default risk

A first strand studies how sovereigns choose between domestic and foreign creditors when default is selective. Broner, Martin, and Ventura (2010) show that secondary-market trading between foreigners and domestic residents can sustain cross-border lending even when sovereigns would otherwise prefer to default on foreigners. Gennaioli, Martin, and Rossi (2014) model the interaction between sovereign default and domestic financial development, showing how the holdings of sovereign debt by domestic banks determine the government’s default incentives. Empirically, Arslanalp and Tsuda (2014) construct a database that tracks global demand for advanced-economy sovereign debt by holder type, and Fang, Hardy, and Lewis (2025) document the patterns and macroeconomic

consequences of sovereign-debt holdings across countries. Our paper builds on this empirical infrastructure but goes further by providing a unified framework for the *three-way* allocation—domestic non-banks, foreigners, and banks—that the data actually display.

2.1.2 Financial repression and domestic debt bias

A second strand studies why governments push debt onto the domestic banking sector. Reinhart and Sbrancia (2015) provide historical evidence that financial repression—a mix of interest-rate ceilings, reserve requirements and captive domestic audiences for government debt—has been a quantitatively important mechanism through which governments liquidate their debt burdens. Chari, Dovis, and Kehoe (2020) rationalize these policies in a Ramsey framework and show that forcing banks to hold sovereign debt can be part of an optimal taxation scheme when the government lacks commitment. Most directly related to our paper, Jeanne (2025) models optimal financial repression as a sequential process: below a debt threshold banks act as marginal buyers (balance-sheet repression), and above it the government extracts quasi-fiscal revenue from the banking sector. That paper establishes why banks must absorb debt beyond a certain level, but treats the non-bank sector as a single passive residual. We take the balance-sheet-repression mechanism as a starting point and ask a complementary question: *among the non-bank creditors, how does debt distribute between domestic non-banks and foreigners, and what determines the cross-country heterogeneity in this allocation?*

2.1.3 Sovereign–bank doom loop

A third strand emphasizes the two-way feedback between sovereign risk and bank balance sheets. Brunnermeier, Garicano, Lane, Pagano, Reis, Santos, Thesmar, Van Nieuwerburgh, and Vayanos (2016) formalize the “diabolic loop” between weak sovereigns and weak banks and propose safe-asset designs that would break it. Farhi and Tirole (2018) develop a general-equilibrium model in which bank bailouts and sovereign distress reinforce each other, generating a genuine doom loop. This literature motivates a central premise of our framework: placing marginal debt in the banking sector is not costless, because bank-held sovereign debt distorts bank balance sheets and amplifies macro-financial feedbacks. Our model captures this by treating bank absorption as the *residual* margin with a convex welfare cost, rather than as a perfectly elastic funding source.

2.1.4 Optimal government debt level

A fourth strand provides a rationale for why non-banks hold government debt in the first place. Aiyagari and McGrattan (1998) show that in an incomplete-markets setting, households value government bonds as a self-insurance instrument against uninsured idiosyncratic income risk, pinning down an optimal positive quantity of public debt. This provides a microeconomic rationale for a downward-sloping, capacity-bounded non-bank demand curve. That framework, however, is closed-economy and has no sovereign default, so it leaves unaddressed the two margins at the center of our paper: sovereign

default risk and the role of foreign investors. Our model does not micro-found the non-bank demand curve in the Aiyagari–McGrattan sense; instead, we posit a reduced-form demand $d_h = S_h \cdot g(r_h)$ consistent with that tradition, and focus on how non-bank demand, a foreign no-default ceiling, and bank residual absorption jointly determine the allocation of sovereign debt.

2.2 Organization

The paper is organized as follows. Section 3 presents the three-sector cost-minimization model, in which the government allocates debt across domestic non-banks with country-specific capacity, foreigners with a uniform no-default ceiling, and banks as the residual absorber with convex financial-repression costs, and derives the resulting equilibrium predictions for debt composition. Section 4 takes the model to the data: we describe the panel of advanced-economy debt-composition data and structural sector-size variables, implement a two-stage estimator that identifies structural sector sizes in a first stage and the cross-country allocation of debt in a second stage, and document the resulting heterogeneous pecking order and concavity in debt absorption. Section 5 concludes.

3 Model

A cost-minimizing government chooses how to allocate a given stock of debt across three creditor sectors—domestic non-banks, foreigners, and the banking sector. The three sectors differ in *how* placement becomes costly at the margin, and those differences produce a tractable pecking order that we take to the data.

3.1 Setting

A government has a given amount of outstanding debt $D > 0$ and chooses non-negative holdings (d_h, d_f, d_b) satisfying

$$d_h + d_f + d_b = D, \tag{1}$$

where d_h is non-bank holdings, d_f is foreign holdings, and d_b is banking (central bank plus private banks).

Every placement is costly.

- **Non-banks and foreigners** lend at interest rates $r_h(d_h, S_h)$ and $r_f(d_f)$; the government’s cash cost of placement is the interest bill $r_h d_h + r_f d_f$.
- **Banking** is not summarised by an interest rate: placing d_b with the banking system carries a separate welfare (quasi-fiscal) cost $\Psi(d_b)$ —inflation, monetary/fiscal trade-offs, financial-stability risk.

Structural size. $S_h > 0$ is the *country-specific* non-bank sector size (capacity).

Planner problem. Total debt $D > 0$ is given. Choose $d_h, d_f, d_b \geq 0$ to minimise total placement cost subject to the budget constraint:

$$\min_{d_h, d_f, d_b} r_h(d_h, S_h) d_h + r_f(d_f) d_f + \Psi(d_b) \quad \text{s.t.} \quad d_h + d_f + d_b = D. \quad (2)$$

3.2 The three sectors: how placement becomes costly

Marginal cost rises with the sector's *degree of saturation*. What differs is *how* saturation is measured.

Non-banks.

$$r_h(d_h, S_h) = \bar{r}_h + \gamma_h \frac{d_h/S_h}{\bar{\omega} - d_h/S_h}. \quad (3)$$

Saturation is the *relative amount* d_h/S_h . The rate stays near $\bar{r}_h$ when the sector is far from its cap $\bar{\omega}S_h$ and diverges as $d_h \rightarrow \bar{\omega}S_h$. Larger S_h means more headroom and later saturation. S_h differs across countries.

Foreigners.

$$r_f(d_f) = \bar{r}_f + \gamma_f \frac{d_f}{\bar{D}_f - d_f}. \quad (4)$$

Saturation is the level of foreign exposure d_f . The rate diverges as $d_f \rightarrow \bar{D}_f$ (a no-default ceiling). $\bar{D}_f$ is *common* across countries.

Banking.

$$\Psi(d_b), \quad \Psi'(d_b) > 0, \quad \Psi''(d_b) > 0. \quad (5)$$

There is no hard ceiling, but placement carries a welfare cost that is increasing and convex.

3.3 Equilibrium: who absorbs the next dollar?

At the optimum, the planner equates the marginal cost of placing one more dollar of debt across all three sectors:

$$MC_h(d_h; S_h) = MC_f(d_f) = MC_b(d_b) = \lambda(D), \quad d_h + d_f + d_b = D. \quad (6)$$

If one sector were cheaper at the margin, the planner would shift toward it.

How the equilibrium plays out as D grows.

- **Low D .** Non-banks and foreigners are far from their limits; MC_h and MC_f are low and they absorb most new debt. Banking is costlier at the margin and absorbs little.

- **Rising D .** Non-banks and foreigners approach their limits; MC_h and MC_f rise sharply and their absorption share shrinks. Banking still has room and starts picking up.
- **High D .** Non-banks and foreigners are near their caps; $MC_h, MC_f \rightarrow \infty$ and they absorb almost nothing at the margin. Banking is the only sector with a finite marginal cost and absorbs nearly all new debt.

3.4 Model predictions

1. **Every sector absorbs more debt as D grows.** Upward-sloping MC_j implies that as D rises, all three margins rise together and every d_j^* rises.
2. **Larger non-bank sectors absorb more debt at any given D .** A bigger S_h means a less saturated non-bank sector at any d_h , a lower MC_h at the margin, and so the planner allocates more to non-banks.
3. **As D grows, non-banks and foreigners absorb a smaller share of each extra dollar; their marginal shares go to zero. Under symmetry, the paths of d_h^* and d_f^* are strictly concave in D .** Both approach caps, so MC_h and MC_f rise and diverge, and the planner shifts each extra dollar toward banking.
4. **Banking takeover at high D .** With non-banks and foreigners near their caps, banking is the only sector with room, and its share of new issuance rises sharply.

3.5 Mechanics I: three phases of absorption

One country, $S_h = 150$. Baseline calibration $\bar{r}_h = \bar{r}_f = \bar{r}_b = 1$, $\gamma_h = \gamma_f = 0.02$, $\bar{\omega} = 0.30$, $\psi = 0.04$, $\bar{D}_f = 55$. Figure 1 shades the three equilibrium phases, defined by banking's marginal share $s_b \equiv dd_b^*/dD$.

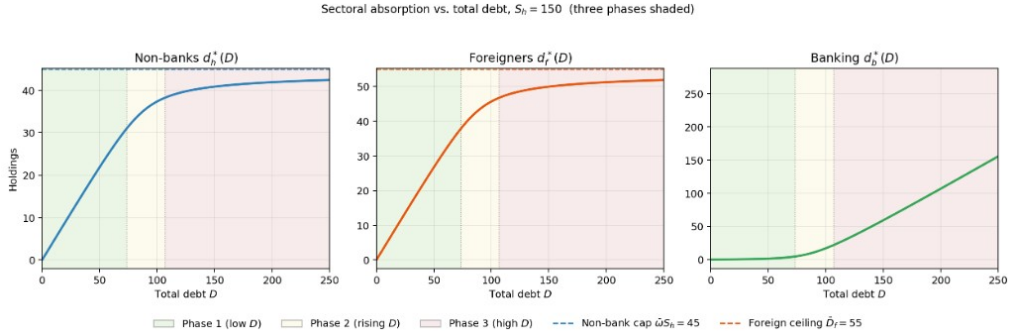


Figure 1 Three phases of debt absorption for a representative country ($S_h = 150$).

- **Phase 1.** Non-banks and foreigners absorb; banking flat.

- **Phase 2.** Non-banks and foreigners saturate; banking starts picking up.
- **Phase 3.** Banking takes over. The transition happens at $D \approx \bar{\omega}S_h + \bar{D}_f = 100$, matching the empirical kink at 100% of GDP documented in Section 4.

3.6 Mechanics II: heterogeneous non-bank sector

Comparative statics in S_h ($S_h \in \{30, 70, 110, 150, 200\}$; $\bar{D}_f, \bar{\omega}, \gamma_h, \gamma_f, \psi$ fixed at the baseline).

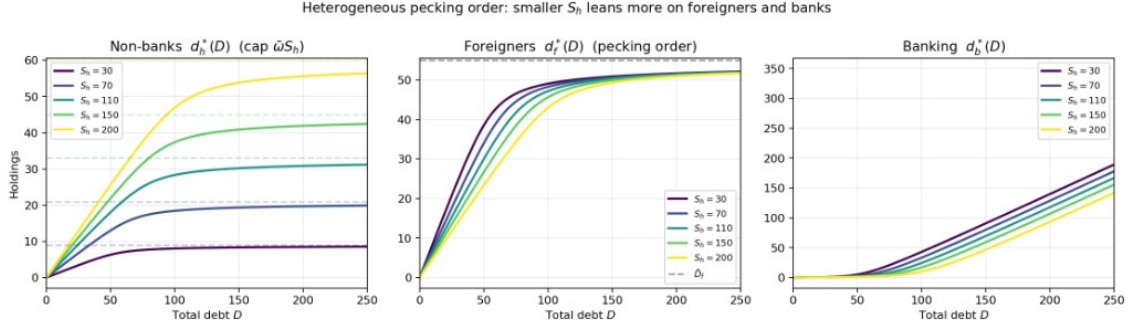


Figure 2 Pecking order across countries with different non-bank sector sizes S_h .

Pecking order.

- **Non-banks:** five different saturation levels, one per country, at $\bar{\omega}S_h$.
- **Foreigners:** essentially one curve—all countries approach the *same* ceiling $\bar{D}_f$, differing only in how fast they get there.
- **Banking:** once total debt is large enough, banking takes over for all countries.

4 Empirical Analysis

We bring the cost-minimizing allocation model of Section 3 to a panel of 32 advanced economies over 1989–2022. The empirical strategy is a two-stage estimator: a first stage recovers each country’s structural sector size $\hat{\delta}_{ij}$, and a second stage uses these as interaction variables to estimate the slope and threshold behaviour of debt allocation.

4.1 Data and sector-size variables

Sources.

- **Debt holdings:** Arslanalp and Tsuda (2014), sovereign debt by creditor sector.

- **Sector-size proxies:** World Bank, *Global Financial Development Database* (GFDD).
- **Panel:** 32 advanced economies, 1989–2022, for a total of $N = 780$ country–year observations. Coverage thickens after 2000; the full 32-country panel is available from 2007 onward.

Country–year proxies for structural sector size S_j .

$$\begin{aligned}
s_{h,it} &= \frac{\text{Pension fund assets}_{it} + \text{Insurance assets}_{it}}{\text{GDP}_{it}} && \rightarrow S_h \text{ (non-bank capacity)} \\
s_{pb,it} &= \frac{\text{Bank deposit funding}_{it}}{\text{GDP}_{it}} && \rightarrow S_{pb} \text{ (private-bank size)} \\
s_{cb,it} &= \frac{\text{Central bank assets}_{it}}{\text{GDP}_{it}} && \rightarrow S_{cb} \text{ (central-bank size)}
\end{aligned}$$

Foreign sector. No country-specific size proxy is available—foreign creditors are global balance sheets, not a local sector we can size in GFDD. We therefore run *no first-stage regression for foreigners*. Foreign debt is treated as the accounting residual

$$d_{f,it} = D_{it} - d_{h,it} - d_{cb,it} - d_{pb,it}. \quad (7)$$

4.2 Empirical strategy: two-stage estimator

Threshold. $\bar{D} = 100\%$ of GDP, chosen from the data: banking’s marginal absorption accelerates sharply once D_{it} crosses this level (discussed in the Motivation).

Stage 1—structural sector size $\hat{\delta}_{ij}$. For each country i and sector $j \in \{h, cb, pb\}$, regress sector share on total debt with country and year fixed effects, allowing a kink at $\bar{D}$:

$$s_{ijt} = \delta_{ij} + \delta_{jt} + \alpha_{1j}D_{it} + \alpha_{2j}\mathbf{1}\{D_{it} \geq \bar{D}\}(D_{it} - \bar{D}) + u_{ijt}. \quad (8)$$

We read $\hat{\delta}_{ij}$, the country fixed effect, as the part of country i ’s sector- j size that does not move with time, after stripping out variation explained by D_{it} and the kink.

Stage 2—debt allocation. The effective slope is a product of a baseline rate and a size scaling, with a kink at $\bar{D}$. In its original factored form,

$$d_{it}^j = \underbrace{\beta_j \left(1 + \gamma_j \hat{\delta}_{ij}\right)}_{\text{baseline slope}} D_{it} + \underbrace{\eta_j \left(1 + \rho_j \hat{\delta}_{ij}\right)}_{\text{post-threshold slope change}} (D_{it} - \bar{D}) + \varepsilon_{ijt}. \quad (9)$$

Expanding the two products gives the estimating equation reported in the tables:

$$d_{it}^j = \beta_j D_{it} + \theta_j (\hat{\delta}_{ij} \cdot D_{it}) + \mu_j \mathbf{1}_{it}(D_{it} - \bar{D}) + \nu_j \mathbf{1}_{it}(\hat{\delta}_{ij} \cdot (D_{it} - \bar{D})) + \varepsilon_{ijt}, \quad (10)$$

with $\theta_j = \beta_j \gamma_j$, $\mu_j = \eta_j$, and $\nu_j = \eta_j \rho_j$.

- β_j : baseline slope—sector j 's absorption per unit of D .
- θ_j : *size-scaling*—larger $\hat{\delta}_{ij}$ implies steeper absorption.
- μ_j, ν_j : post-threshold slope change and its size-scaling.

4.3 First-stage results: structural sector sizes

Table 1 reports estimates of (8).

Table 1 First stage: structural sector sizes

	(1) Non-bank s_{iht}	(2) Central bank s_{icbt}	(3) Private bank s_{ipbt}
$\alpha_{1j}: D_{it}$	−0.203 (0.234)	0.016 (0.044)	−0.321 (0.265)
$\alpha_{2j}: \text{highD}(D_{it} - \bar{D})$	−0.184 (0.203)	0.520*** (0.167)	0.856** (0.372)
N	547	718	732
Groups	31	32	32
R^2 (within)	0.436	0.603	0.199
Year / Country FE	Yes	Yes	Yes

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Takeaway.

- **Non-banks.** α_{1h}, α_{2h} are insignificant: non-bank size does not move with total debt, consistent with a *hard* capacity $\bar{\omega}S_h$.
- **Central- and private-bank size.** $\alpha_{2j} > 0$ and significant: banking-sector size *grows* once $D_{it} > \bar{D}$, consistent with banking's no-hard-ceiling property in the model.

4.4 Second-stage results: debt allocation

Table 2 reports estimates of (10).

Takeaways.

- **All sectors absorb part of new debt.** $\beta_j > 0$ across the three sectors.

Table 2 Second stage: debt allocation by sector

	Non-bank (d_h)	Central bank (d_{cb})	Private bank (d_{pb})
$\beta_j: D_{it}$	0.314*** (0.040)	0.053*** (0.012)	0.192*** (0.020)
$\theta_j: \hat{\delta}_{ij} \cdot D_{it}$	0.0018*** (0.0005)	0.0131*** (0.0044)	-0.0002 (0.0007)
μ_j : threshold slope	-0.018 (0.171)	0.334** (0.153)	0.462** (0.220)
ν_j : threshold $\times \hat{\delta}$	-0.004 (0.003)	-0.055 (0.036)	-0.009 (0.008)
N	753	780	780
F	83.26	177.55	72.93
R^2	0.802	0.705	0.819

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

- **Non-banks scale with size.** $\theta_h > 0$ and significant: countries with intrinsically larger non-bank sectors absorb a larger share of government debt at any given D . This maps to model prediction 2 (size scaling).
- **Banking takes over above $\bar{D}$.** $\mu_{cb} + \mu_{pb} \approx 0.80$: banking's marginal absorption jumps by roughly 80 percentage points once $D_{it} > 100\%$ of GDP. This maps to model prediction 4 (banking takeover).

Foreign debt residual. Using the fitted domestic components from (10), the implied foreign residual tracks the data: correlation = 0.779, slope = 0.744***, $R^2 = 0.608$, MAE = 8.79.

4.5 Concavity in debt absorption

Specification. For sector $j \in \{h, f\}$,

$$d_{it}^j = \beta_j D_{it} + \kappa_j D_{it}^2 + \theta_j (\hat{\delta}_h \cdot D_{it}) + \varepsilon_{it}, \quad (11)$$

where θ_j is included only for the non-bank sector. Pooled and country-FE estimates are reported separately, with and without $\hat{\delta}_h \cdot D$.

Table 3 reports estimates of (11).

Takeaways.

- **Non-banks.** Concavity ($\kappa_h < 0$) shows up *within country* but disappears once we control for size ($\hat{\delta}_h \cdot D$): the slowdown is driven by country-specific capacity S_h .

Table 3 Concavity in debt absorption

	Non-bank (d_h)				Foreign (d_f)	
	Spec C: no $\hat{\delta}_h$		Spec B: with $\hat{\delta}_h \cdot D$		No $\hat{\delta}_h \cdot D$	
	Pooled	FE	Pooled	FE	Pooled	FE
$\beta(D)$	0.343***	0.402***	0.359***	0.299*	0.387***	0.392***
$\theta(\hat{\delta}_h \cdot D)$	—	—	0.0022***	0.0011*	—	—
$\kappa(D^2)$	−0.0003	−0.0016***	−0.0010	−0.0009	−0.0014***	−0.0009
N	780	780	701	701	747	747
R^2	0.735	0.381	0.801	0.381	0.773	0.454

Notes: Standard errors omitted for space. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

- **Foreigners.** Concavity ($\kappa_f < 0$) shows up *across countries* but not within: consistent with a *common* foreign ceiling $\bar{D}_f$.

Map to predictions. These two findings jointly support model prediction 3: the marginal absorption shares of non-banks and foreigners fall as D grows, but the source of concavity differs by sector—country-specific S_h for non-banks, a common $\bar{D}_f$ for foreigners.

5 Conclusion

Most of the sovereign-debt literature focuses on the *level* of government debt: how much a country can sustain, and at what cost. This paper has shifted the question to the *allocation* of that debt across holder types. The data on advanced economies show that countries with nearly identical debt-to-GDP ratios place their debt in very different hands—some lean on domestic pension funds and insurers, others on foreign investors, and others on their banking sector—and that the bank share rises sharply once total debt passes a threshold of roughly 100–120% of GDP. Our paper provides a simple, structural account of this heterogeneous pecking order.

On the *model* side, we studied a cost-minimizing government that allocates debt across three sectors, each with a distinct supply structure: domestic non-banks, whose capacity is tied to a country-specific structural sector size S_h and whose cost rises in the utilization rate d_h/S_h ; foreigners, whose capacity is bounded by a uniform no-default ceiling $\bar{D}_f$ at which the cost of additional borrowing diverges; and banks, which serve as the residual absorber and face a convex welfare cost reflecting crowding-out and financial-repression distortions. The resulting equilibrium is a pecking order whose ordering across sectors depends on country-specific parameters rather than being universal.

On the *empirical* side, taking the model to panel data on advanced economies delivers four findings that map one-to-one to the model’s predictions. First, cross-country heterogeneity in debt allocation is driven primarily by the structural size of the non-bank sector, not by the total debt level or by openness alone. Second, the banking sector is the elastic margin: banks absorb the residual after non-bank and foreign capacities are exhausted, which is why the bank share accelerates above the threshold. Third, foreign absorption exhibits the concavity consistent with a uniform no-default ceiling—foreign holdings rise with debt but at a decreasing rate, with a common capacity limit across countries. Fourth, financial integration has only a modest additional effect: more open economies place somewhat less domestic government debt with non-banks, but the effect is second-order relative to the structural sector-size channel.

These results carry a simple message for the policy debate on sustainable debt levels. Two countries with the same debt-to-GDP ratio can face very different sustainability pressures depending on the structural size of their non-bank financial sector and their distance to the foreign no-default ceiling. Countries with large, deep non-bank sectors can absorb more debt at home without triggering the convex repression cost on banks, while countries with structurally small non-bank sectors hit that margin sooner. Our framework does not attempt to micro-found the non-bank demand curve in the Aiyagari and McGrattan (1998) tradition, nor to endogenize sovereign default in the full Broner, Martin, and Ventura (2010) or Gennaioli, Martin, and Rossi (2014) sense; these extensions, together with richer dynamics of how $\bar{D}_f$ responds to shocks, are natural next steps. Even so, the tractable three-sector structure developed here provides a transparent lens through which to interpret cross-country and over-time variation in who holds government debt, and a portable empirical design for future work on sovereign-debt composition.

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